longer you wait, the more likely it is that there'll be some random fluctuation in the electronics so strong that it generates a spurious signal. So we ignore anything that isn't *much* louder than the background.

Any strong narrow-band signal that remains in a single channel we take very seriously. As it logs in the data, META automatically tells the human operators to pay attention to certain signals. Over five years we made some 60 trillion observations at various frequencies, while examining the entire accessible sky. A few dozen signals survive the culling. These are subjected to further scrutiny, and almost all of them are rejected—for example, because an error has been found by fault-detection microprocessors that examine the signal-detection microprocessors.

What's left—the strongest candidate signals after three surveys of the sky—are 11 "events." They satisfy all but one of our criteria for a genuine alien signal. But the one failed criterion is supremely important: Verifiability. We've never been able to find any of them again. We look back at that part of the sky three minutes later and there's nothing there. We look again the following day: nothing. Examine it a year later, or seven years later, and still there's nothing.

It seems unlikely that every signal we get from alien civilizations would turn itself off a couple of minutes after we begin listening, and never repeat. (How would they know we're paying attention?) But, just possibly, this is the effect of twinkling. Stars twinkle because parcels of turbulent air are moving across the line of sight between the star and us. Sometimes these air parcels act as a lens and cause the light rays from a given star to converge a little, making it momentarily brighter. Similarly, astronomical radio sources may also twinkle—owing to clouds of electrically charged (or "ionized") gas in the great near-vacuum between the stars. We observe this routinely with pulsars.

Imagine a radio signal that's a little below the strength that we could otherwise detect on Earth. Occasionally the signal will by chance be temporarily focused, amplified, and brought within the detectability range of our radio telescopes. The interesting thing is